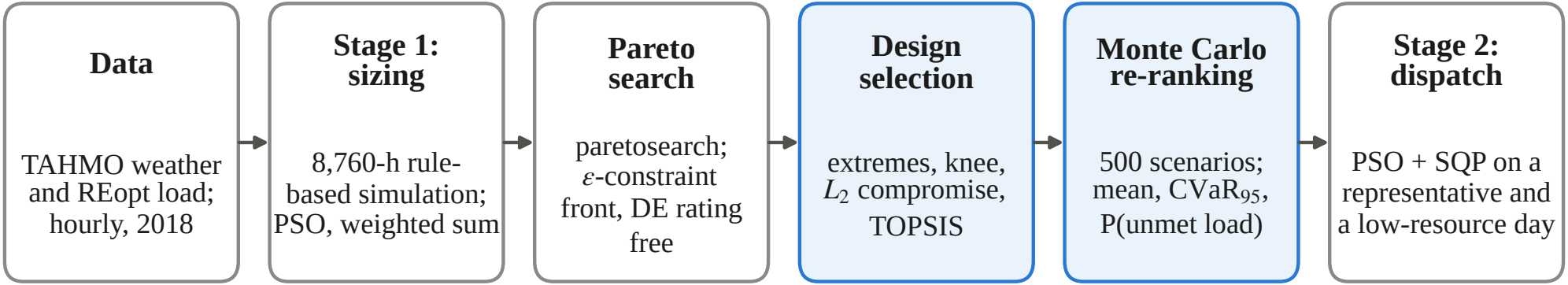
Data
TAHMO weather and REopt load; hourly, 2018
Stage 1: sizing
8,760-h rule-based simulation; PSO, weighted sum
Pareto search
paretosearch; ε-constraint front, DE rating free
Design selection
extremes, knee, L_2 compromise, TOPSIS
Monte Carlo re-ranking
500 scenarios; mean, $CVaR_{95}$, P(unmet load)
Stage 2: dispatch
PSO + SQP on a representative and a low-resource day